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From Freshwater to High Saline Produced Water: Unlocking the Potential of Hydraulic Fracturing in Water-Scarce South Oman

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Abstract

The scale-up of multistage hydraulic fracturing in South Oman has intensified challenges related to sourcing sufficient fresh water. With approximately 8,000 m³ of water demand per well, the reliance on water haulage has led to extended standby times, increased operational costs, and production delays. This study explores the feasibility of using locally available high saline produced water—previously considered unsuitable for fracturing—to enhance operational efficiency while addressing freshwater scarcity.

A comprehensive evaluation was conducted on various alternative water sources, including outputs from aquifer pump-off projects, wetlands, reedbeds, and deep-water wells, all were assessed based on volumetric capacity and logistical feasibility. Each source was assessed in laboratory to provide detailed chemical composition, and conduct scaling risks evaluation under simulated reservoir, production, and surface conditions when mixing with reservoir fluids. The study further explored the use of alternative chemical agents formulated to tolerate impurities, enabling the integration of these high-salinity water sources into fracturing operations.

Initial compatibility tests identified different scaling risks in several water sources, leading to the selection of the most promising candidate for further evaluation. The standard slickwater formulations with high-viscosity friction reducers (HVFR) normally used for these multistage frac operations, were found to exhibit compromised rheological properties when mixed with the selected high saline water, thereby affecting proppant placement. This occurs because, the hydration of the polyacrylamide-based additive is compromised by the salinity of the mixing water. An alternative HVFR additive, specifically engineered for high salinity and impurity tolerance, was subsequently evaluated and approved. A field trial during the final stage of a well confirmed the operational viability of replacing fresh water with the produced water in multistage fracturing operations. The implementation of this technology has great benefits for the company and the country in terms of cost savings due to logistic and operations and allocating fresh water for human consumption only.

Introduction

Field and Reservoir Description

The ALN field in Southern Oman is known for its challenging ATL silicilite formation—a tight, laminated, and partially shaly rock encased within thick salt intervals. The deposit is located at 3 km deep formation which is primary silica, is a unique source and reservoir rock. The rock is organic rich, finely layered, has high porosity (Up to 30%) but low permeability (~ 0.01 md). Net pay thickness varies from approximately 250 m in flank areas to 350 m in the crest.

This formation is one of the most technically challenging and operationally constrained tight gas reservoirs within the Southern Oman Basin. It is a rock with low permeability and high sour gas content. After 24 years of production and depletion from earlier completions, infill stimulation becomes essential to access bypassed reserves and sustain reservoir pressure via gas injection.

This paper documents the multi-stage hydraulic fracturing methodology implemented on a gas injector well, focusing on novel HVFR fluid systems compatible with high salinity water to manage closely optimization of water resources and ultimately the cost of the well completion and workover. The results of the technical, operational and logistics efforts unlock broader opportunity in the field to maximize the recovery.

Statement of Theory and Definitions

Functions and Properties of Fracturing Fluids in Hydraulic Stimulation

The primary functions of a fracturing fluid are to create and extend the fracture, transport proppant through the mixing and pumping equipment and into the fracture and place the proppant at the desired location in the fracture. Failure to perform any of these functions effectively may compromise the stimulation treatment. To achieve optimal performance, fracturing fluids should exhibit the following properties:

- Low tubular friction losses to minimize horsepower requirements.
- Minimal fluid loss to the formation (high fluid efficiency, defined as the fraction of retained fluid versus total pumped volume).
- Effective proppant transport capacity to ensure uniform distribution.
- Controlled viscosity to maximize fracture penetration while maintaining sufficient width for proppant placement.
- Residue-free breakdown to avoid proppant-pack damage.
- Compatibility with reservoir fluids, rock, at in-situ temperature/pressure conditions.
- Accessibility of base fluids for cost-effective preparation.
- Safety and environmental compliance to mitigate hazards.
- Economic viability with minimized preparation costs.

Slickwater Fracturing in Low-Permeability Formations

A successful hydraulic fracturing stimulation fully distributes the proppant through the conductive fracture networks, a process that strongly depends on choosing the appropriate fracture fluid type and proppant size. In very low-permeability reservoirs, like ATL formations, successful stimulation requires long fractures with low proppant concentrations, often forming partial monolayers. Slickwater fracturing—employing High-Viscosity Friction Reducers (HVFRs)—is the preferred technique. This method uses large water volumes pumped at high rates, with HVFR additives to reduce friction and enhance proppant suspension. However,

key challenges include water availability for high consumption operations and compatibility of friction reducers with local water chemistry.

The most common HVFR agents used with water for this kind of operations are partially hydrolyzed acrylamide polymer. The polyacrylamide polymers are generally used as oil-external emulsions with either a built-in inverter or an additional inverter that flips the emulsion and hydrates the polymer, giving some viscosity and reduction in friction properties. These polymers develop sufficient viscosity for proppant transport under turbulent conditions. Compared to conventional fluids (e.g., crosslinked guar gels), HVFRs offer lower operational costs for equivalent fracture conductivity and superior proppant transport under turbulent flow.

Several studies of the rheological properties of HVFRs, including the viscosity and elasticity of the fluids concluded that HVFR exhibits better viscosity and elasticity compared to the conventional frac fluid polymers, like guar. Also, investigations on the settling velocity of the sand using different fracture fluids showed that HVFR provided longer suspension of the proppant than the guar. The effect of TDS on the friction reducer have also been investigated, to measure the effect of calcium, sodium, potassium, and magnesium brines. The results showed that calcium and magnesium have a greater effect to decrease the friction reducer stability where the percentage of the friction reduction of both brines was lower compared to sodium and potassium.

The effect of the formation impairment due to fracturing fluids residues after addition of breakers has been also investigated using sandstone and shale cores. The outcomes favor the HVFR systems. The regained permeability in sandstone cores reached up to 90% and show minor damage; furthermore, when testing guar based fracturing fluid, some residuals appeared which is considered as one the disadvantages that leads to permeability reduction.

Description of Application and Processes

The standard fracturing fluid system in the ALN field consists of fresh water (sourced from local reverse osmosis (RO) plants) combined with high-viscosity friction reducer (HVFR) additives. Since fresh water is scarce in the region, its availability is a critical logistical factor. Even minor planning deviations can lead to delays, costly non-productive time (NPT) for fracturing units, and deferred production. For instance, an unplanned shutdown of an RO plant during one fracturing campaign resulted in 12 NPT days of the frac unit.

To address this challenge, the company initiated a project to evaluate alternative fluid systems that do not depend on high-quality water. The objective was to utilize locally available non-fresh water sources while ensuring no incompatibilities at reservoir and production conditions, as well as the chemical formulation met the required rheological performance for hydraulic fracturing.

After initial screening, considering volumes availability, based chemical water composition and proximity to the field, 04 different sources of water were selected to be evaluated for the hydraulic fracturing application.

Table 1—Volumes of potential sources of water (m3/year)

Potential source of water in South	2024	2025	2026	2027	2028
APO water	31,892	40,618	47,293	50,348	52,566
Wetlands Treatment Plant	46,000	46,000	45,993	46,000	46,000
Reedbeds water	155,000	155,000	155,000	155,000	155,000
DWD water	28,665	29,170	30,557	32,511	33,892

APO aquifer pump-off water is produced from aquifers in AML field and is used in the steam plant to be injected as secondary recovery in these wells. There is sufficient water being relocated due to some non-

acceptance. This relocation of water represents additional energy consumption and associated cost to the company.

The Wetlands plant uses nature-based solution for treating and disposing of Oil Produced Water (OPW). This plant uses a biological treatment process by harnessing natural processes in wetlands and evaporation zones.

The Reed Beds is another plant used by the company to dispose produced water. This plant used also the wetland technology.

The DWD, is produced water being disposed into deep wells.

Scale assessment of water

The ATL formation does not contain any reservoir water, but H₂S. This fluid reservoir characteristics were considered for the compatibility and risk evaluation. For this, the company conducted a full-scale assessment of the potential sources of water. Representative preserved water samples (adding nitric acid to pH =2 to prevent metals precipitation and microbial activity) were collected and transfer to the lab for full chemistry composition analysis. Lab results are the following:

Table 2—Chemistry composition of water sources

		Water Source			
CATION ANALYSIS		APO	Wetlands	Reed Beds	DWD
Calcium	mg/l	206.6	428.4	518	208.1
Magnesium	mg/l	89.4	114.5	178.7	43.2
Sodium	mg/l	4373	4553	8959	2593
Boron	mg/l	-	44.6	64.4	-
Barium	mg/l	0	0.0997	0.172	0.86
Iron	mg/l	0.33	<0.21	<0.21	0.59
Potassium	mg/l	73.8	74.7	167	40.1
Strontium	mg/l	3	3.77	9.33	2.2
Sulphur	mg/l	-	779	161	93.8
ANION ANALYSIS					
Bicarbonate	mg/l	463	550	324	1240
Carbonate	mg/l	0	0	0	0
Chloride	mg/l	3330	3750	7740	2360
Sulphate	mg/l	500	402	490	110

The self-scale tendency analysis was conducted at different conditions of pressure and temperature: reservoir, bottomhole and surface to ensure no potential risk present during the job and production cycle of the well.

The full scaling tendency of the proposed source water are summarized in Appendix-A. The common scaling (i.e. Barite and Calcite) tendencies are illustrated in [fig-1](#) to [8](#) for the studied sources of water. The Calcite scaling tendency is observed to be directly proportional to temperature for all water sources. Hence, the scaling tendency of the downhole condition is higher than the scaling tendency at the surface condition. However, the severity of scaling is predicted to be low due to the low saturation of the fluids. On the other hand, Barite scaling is observed to be inversely proportional to temperature and scaling tendency over all

is negligible. The selected water source was APO water because it exhibits lower scaling tendencies for common scales and with negligible severity, as illustrated in [fig-9](#).

Additionally, the impact of dissolved H₂S on scaling tendency was studied. The dissolved H₂S is observed to have a positive impact on Calcite and Barite scaling due to acidification phenomena.

Source Water APO

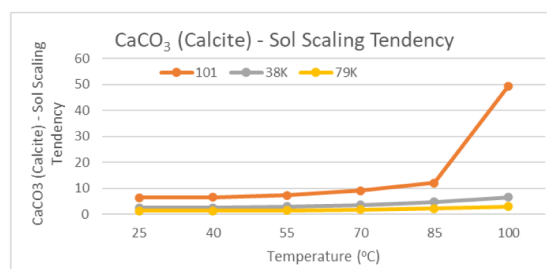


Figure 1—Calcite Scaling tendency for the various temperatures and pressures.

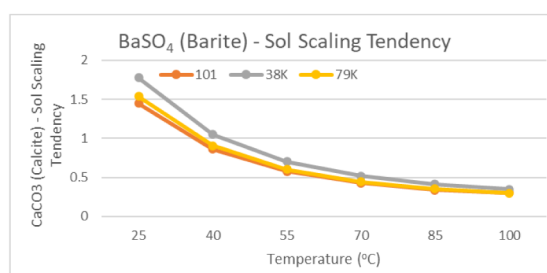


Figure 2—Barite Scaling tendency for the various temperatures and pressures.

DWD

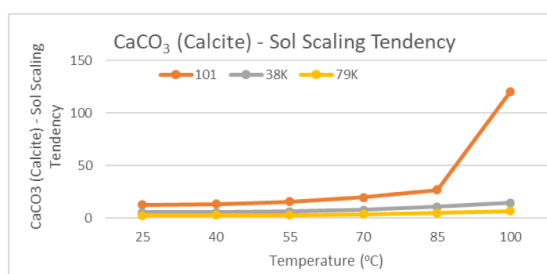


Figure 3—Calcite Scaling tendency for the various temperatures and pressures.

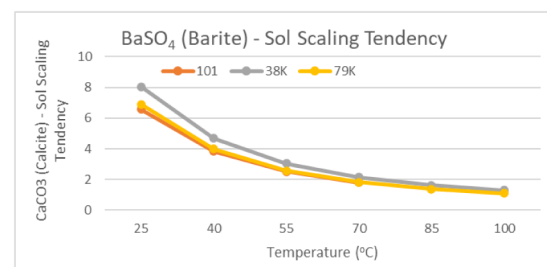


Figure 4—Barite Scaling tendency for the various temperatures and pressures.

Reed Beds

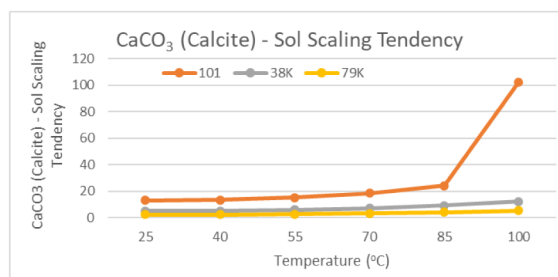


Figure 5—Calcite Scaling tendency for the various temperatures and pressures.

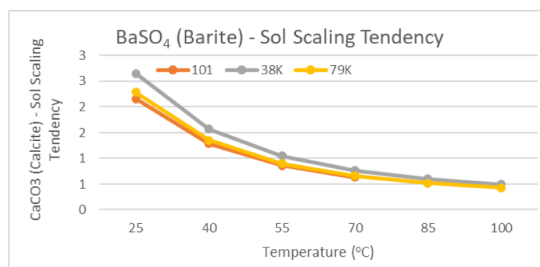


Figure 6—Barite Scaling tendency for the various temperatures and pressures.

Wetlands

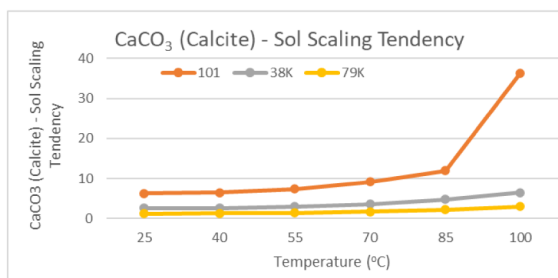


Figure 7—Calcite Scaling tendency for the various temperatures and pressures.

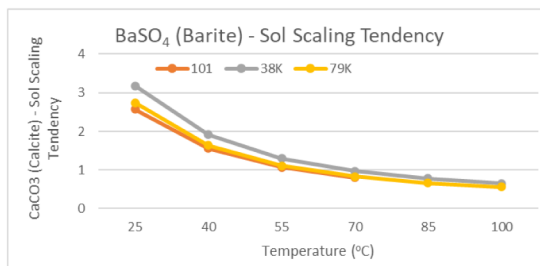


Figure 8—Barite Scaling tendency for the various temperatures and pressures.



Figure 9—Common scale tendencies for all source water.

Frac fluid system evaluation

Following the identification of APO water as the optimal source for the ATL fracturing operation, laboratory validation of the High-Viscosity Friction Reducer (HVFR) system commenced. Five friction reducer formulations from three different vendors were evaluated for their rheological performance and proppant suspension capabilities.

The evaluation was initially done by measuring the HVFR system viscosity at surface temperature. This was the first screening for the system selection and then, the system was evaluated by measuring its viscosity at downhole temperature conditions (100°C) considering the effect of oxidizer breakers.

Of the five products tested, only two met the required viscosity threshold of ≥ 40 cP at 170 s^{-1} at 25°C. Product C1 achieved the target viscosity at a concentration of 12 L/m^3 , whereas Product C2 demonstrated similar performance at a lower dosage of 8 L/m^3 .

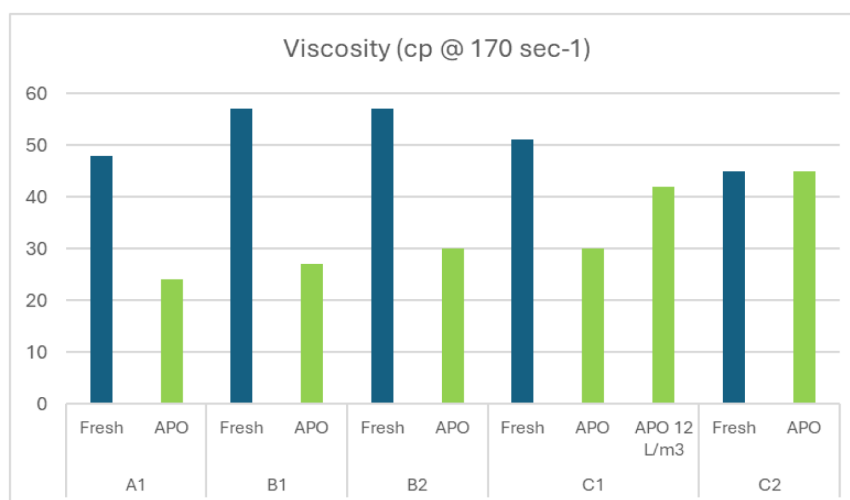


Figure 10—Viscosity of HVFR systems

The impact of oxidizer breakers on the selected HVFR systems was also evaluated to assess their effectiveness in reducing fluid viscosity, thereby enhancing post-fracturing recovery and mitigating formation damage. Breaker dosage was optimized to maintain a minimum viscosity of 25 cP at 100 s⁻¹ under bottomhole static temperature (BHST) conditions of 100°C throughout the pumping operation.

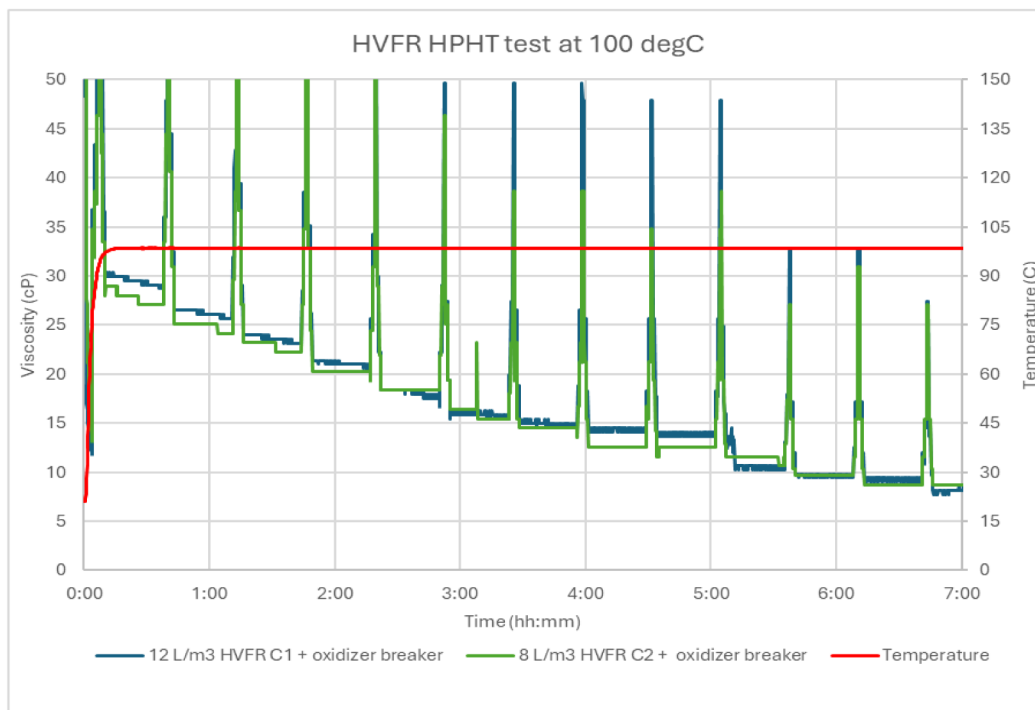


Figure 11—Viscosity of C1 HVFR system @ 100C

Field implementation

The newly developed HVFR system, formulated for compatibility with APO water, was successfully implemented in a three-stage hydraulic fracturing operation within ALN field's 2024 campaign. Due to limited HVFR inventory, the technology was strategically deployed in the final stimulation stage.

During the preparation phase of the job, field sampling revealed elevated turbidity levels (≤ 80 NTU) in the APO water source. To mitigate solids-related risks, a filtration protocol was implemented, involving the deployment of a mobile cartridge filtration unit to achieve target turbidity of <10 NTU (from initial 80 NTU) and to ensure total suspended solids (TSS) compliance.

At the end of 2024, frac unit was mobilized to the selected well and preparation initiated. A total of 3 frac stages were conducted with no disruption due to logistic issues (like the ones experienced in the past). The last frac stage was conducted using the new HVFR high TDS tolerant system. A total of 84.5 MT proppant was pumped into formation using 692.7 m³ of slickwater fluid.

At the end of the last proppant concentration, the LAS pump was not able to continue pumping the friction reducer due to an operational issue of the same. Because of this, the viscosity of the system was reduced and early screen-out occurred (90.2% placement).

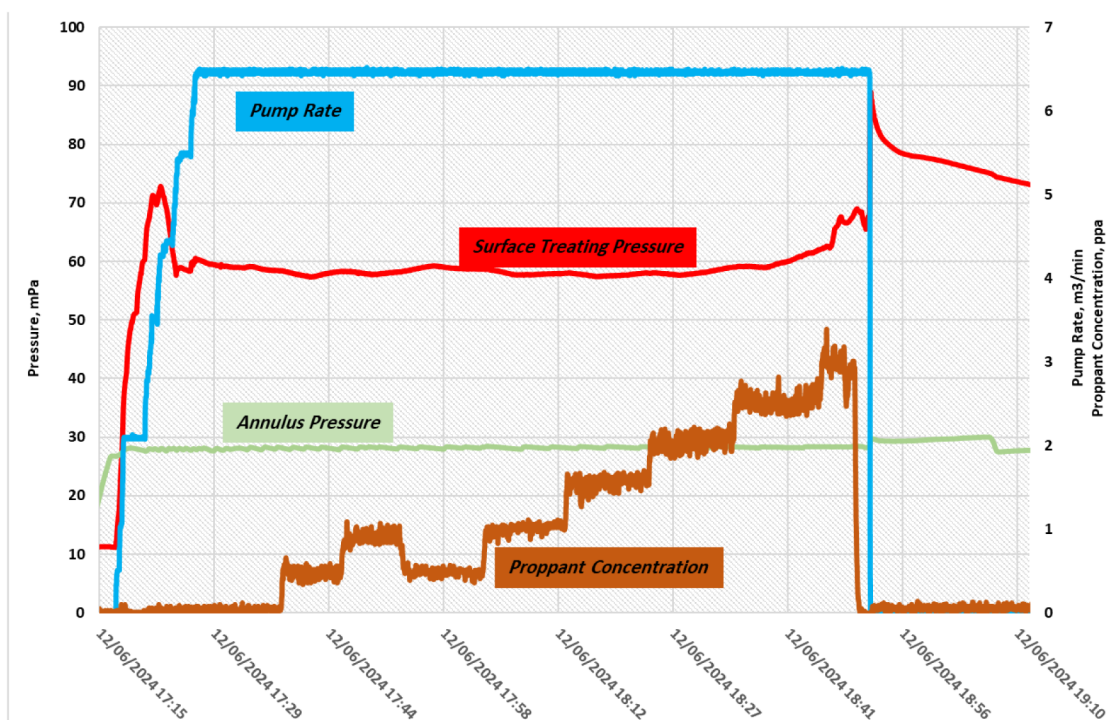


Figure 12—Frac job chart on ALN-42 well

During the pumping operation, the fluid performed as per plan with not deviations on the viscosity behavior. Also, the pumping pressure was under the expected values, confirming the friction reduction effectiveness. Unfortunately, the last 9 MT of proppant pumped in the well, were not placed into the formation due to the malfunction of the LAS system. This was captured as a lesson learned for future jobs.

Conclusions

This work represents the first successful application of high saline produced water in hydraulic fracturing in South Oman. The structured workflow for evaluating alternative water sources, combined with successful field validation, not only demonstrates a viable method to reduce freshwater dependency but also offers significant operational and environmental benefits by minimizing standby time, production delays and reduction on power to inject same water to deep disposal wells. This breakthrough paves the way for more sustainable fracturing practices in water-scarce regions.

Acknowledgement

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Appendix-A

APO water

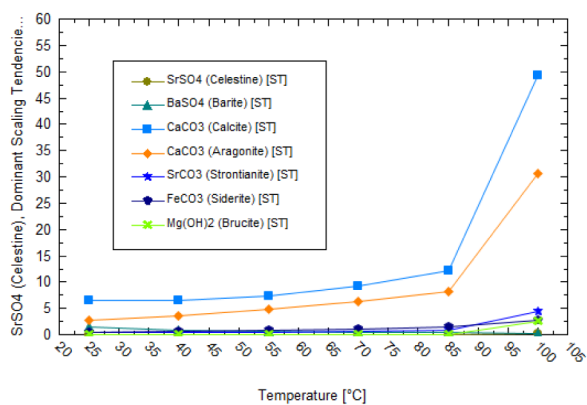


Figure 13—Temperature Survey at 101 kPa (Atmospheric condition)

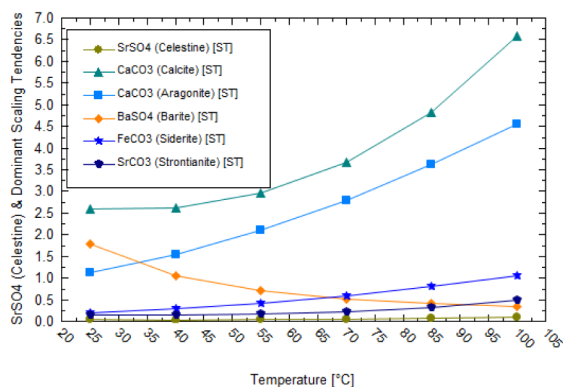


Figure 14—Temperature Survey at 38,000 kPa

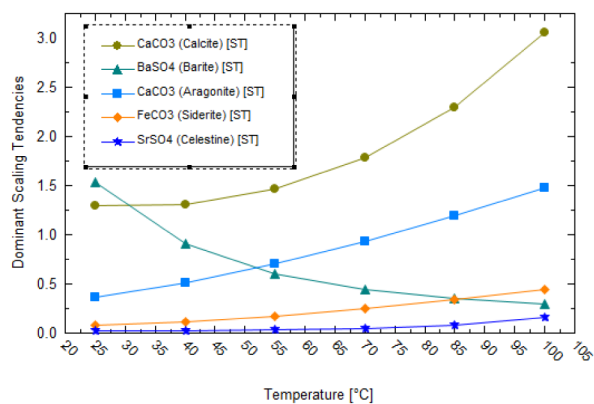


Figure 15—Temperature Survey at 79,000 kPa

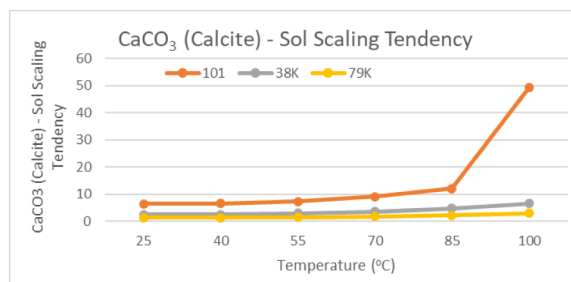


Figure 16—Calcite Scaling tendency for the various temperatures and pressures.

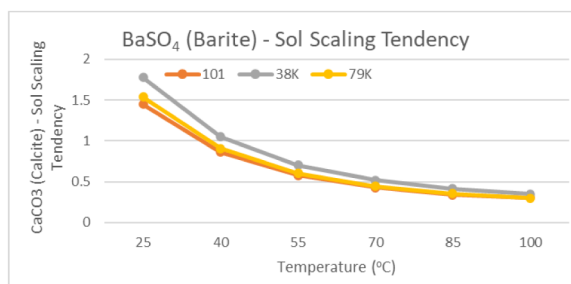


Figure 17—Barite Scaling tendency for the various temperatures and pressures.

DWD water

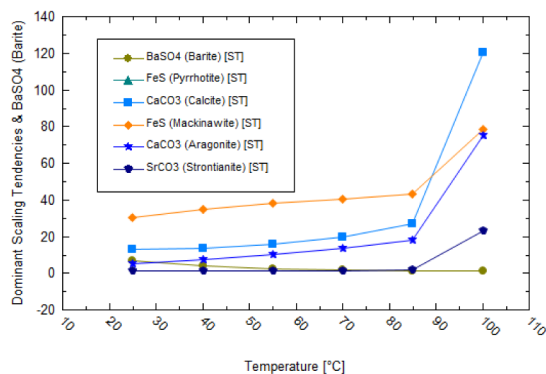


Figure 18—Temperature Survey at 101 kPa (Atmospheric condition)

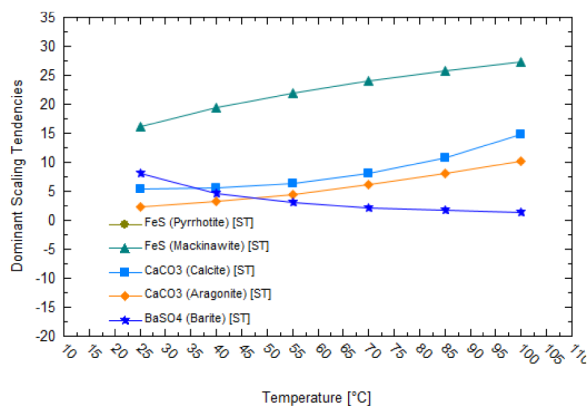


Figure 19—Temperature Survey at 38,000 kPa (Atmospheric condition)

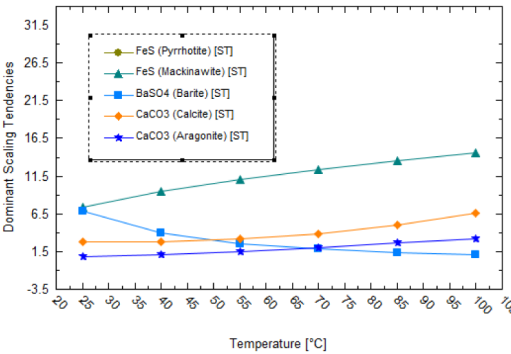


Figure 20—Temperature Survey at 79,000 kPa (Atmospheric condition).

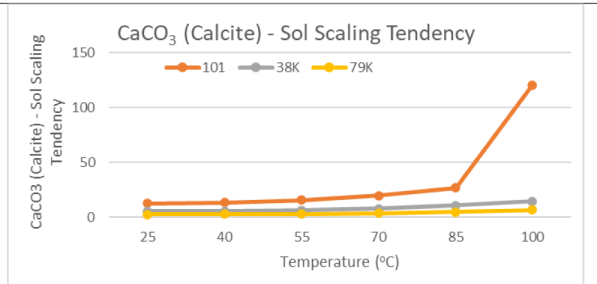


Figure 21—Calcite Scaling tendency for the various temperatures and pressures.

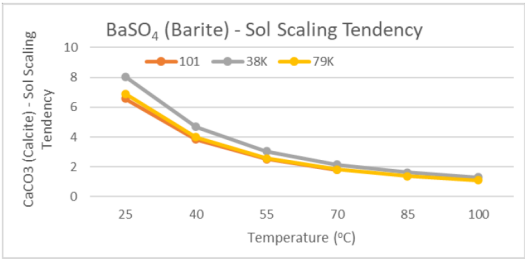


Figure 22—Barite Scaling tendency for the various temperatures and pressures.

Reedbeds water

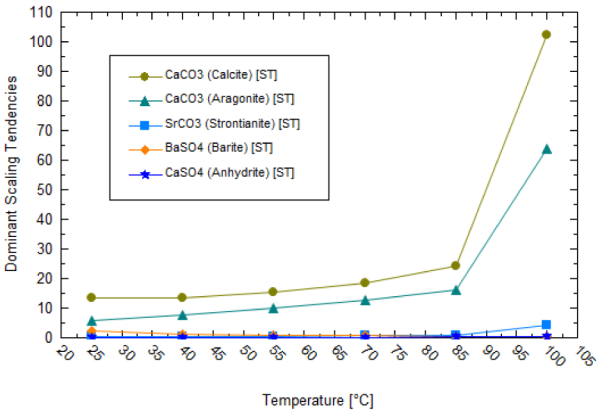


Figure 23—Temperature Survey at 101 kPa (Atmospheric condition).

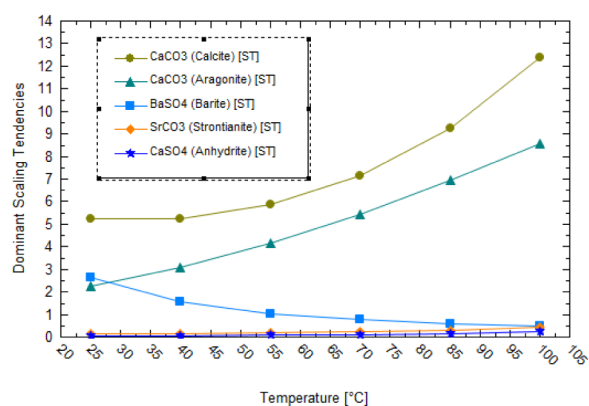


Figure 24—Temperature Survey at 38,000 kPa (Atmospheric condition).

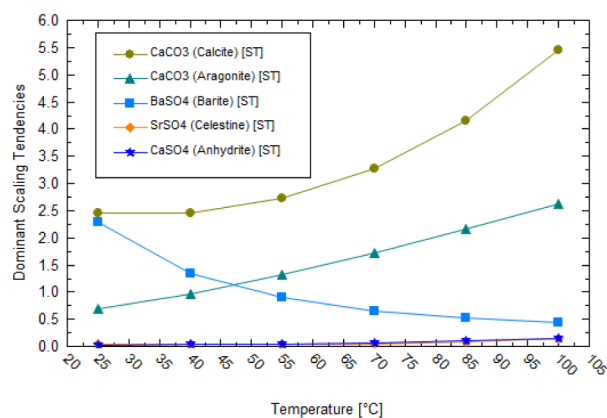


Figure 25—Temperature Survey at 79,000 kPa (Atmospheric condition).

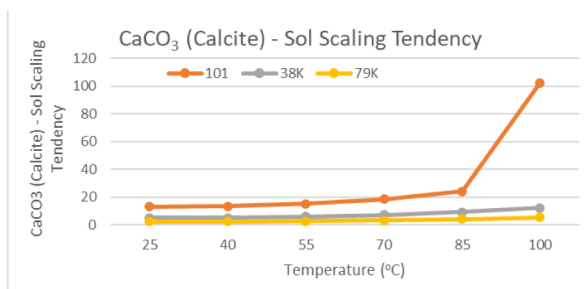


Figure 26—Calcite Scaling tendency for the various temperatures and pressures.

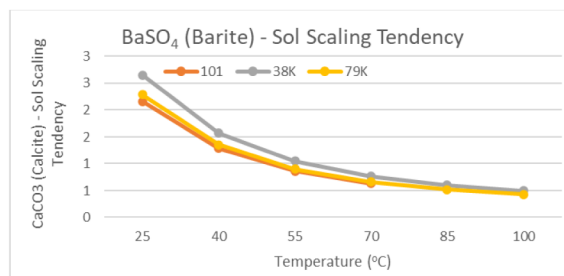


Figure 27—Barite Scaling tendency for the various temperatures and pressures.

Wetlands treatment plant

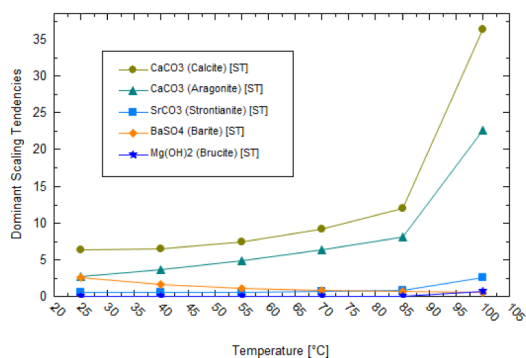


Figure 28—Temperature Survey at 101 kPa (Atmospheric condition).

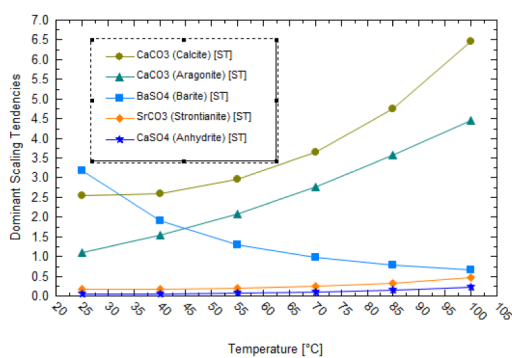


Figure 29—Temperature Survey at 38,000 kPa (Atmospheric condition).

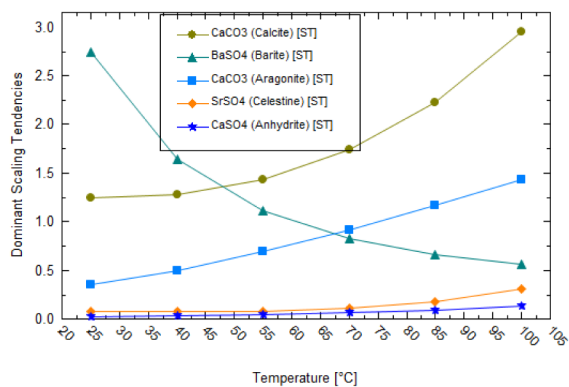


Figure 30—Temperature Survey at 79,000 kPa (Atmospheric condition).

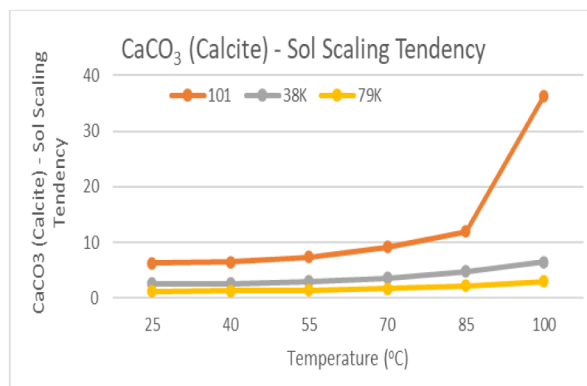


Figure 31—Calcite Scaling tendency for the various temperatures and pressures.

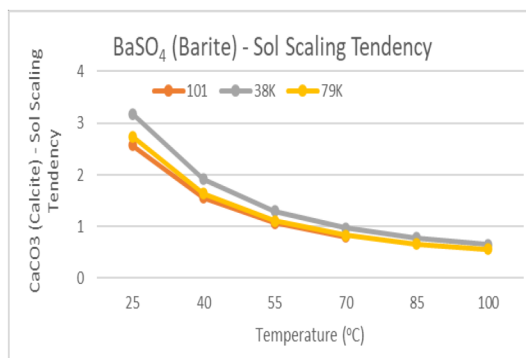


Figure 32—Barite Scaling tendency for the various temperatures and pressures.